

$$1) \left(\frac{dy}{dx}\right)^2 - x \frac{dy}{dx} + y = 0$$

$$p = \frac{dy}{dx}$$

$$p^2 - xp + y = 0$$

$$y = xp - p^2$$

$$\frac{dy}{dx} = x \frac{dp}{dx} + p - 2p \frac{dp}{dx}$$

$$\cancel{p} = x \frac{dp}{dx} + \cancel{p} - 2p \frac{dp}{dx}$$

$$0 = (x - 2p) \frac{dp}{dx}$$

$$p = C$$

$$y = xC - C^2$$

$$y = \frac{x^2}{2} - \frac{x^4}{4}$$

$$4y = 2x^2 - x^4$$

$$4y = x^2$$

$$x - 2C = 0$$

$$x = 2C$$

$$\frac{x}{2} = C$$

$$2) y - x \frac{dy}{dx} = 3 \left(\frac{dy}{dx}\right)^2$$

$$y - xp = 3p^2$$

$$\frac{dy}{dx} = p + x \frac{dp}{dx} + 6p \frac{dp}{dx}$$

$$\cancel{p} = \cancel{p} + x \frac{dp}{dx} + 6p \frac{dp}{dx}$$

$$\frac{dp}{dx} (x + 6p) = 0$$

$$p = C$$

$$y - xC = 3C^2$$

$$y = 3C^2 + xC$$

$$x - 6p = 0$$

$$x = 6p$$

$$x = 6C$$

$$C = -\frac{x}{6}$$

$$y = 3 \left(\frac{x}{6}\right)^2 + x \left(\frac{x}{6}\right)$$

$$y = 3 \left(-\frac{x}{6}\right)^2 + x \left(-\frac{x}{6}\right)$$

$$y = \frac{3x^2}{36} - \frac{x^2}{6}$$

$$y = -\frac{1}{12}x^2$$

$$-12y = x^2$$

$$3) \quad x \left(\frac{dy}{dx} \right)^3 - y \left(\frac{dy}{dx} \right)^2 + 1 = 0$$

$$xp^3 - yp^2 + 1 = 0$$

$$yp^2 = xp^3 + 1$$

$$y = xp + \frac{1}{p^2}$$

$$\frac{dy}{dx} = p + x \frac{dp}{dx} - 2p^{-3} \frac{dp}{dx}$$

$$p = p + x \frac{dp}{dx} - 2p^{-3} \frac{dp}{dx}$$

$$0 = \frac{dp}{dx} (x - 2p^{-3})$$

$$p = c$$

$$y = xc + \frac{1}{c^2}$$

$$x - \frac{2}{p^3} = 0$$

$$x = \frac{2}{p^3}$$

$$x = \frac{2}{c^3}$$

$$y = x^3 \sqrt{\frac{2}{x}} + \frac{1}{\sqrt[3]{\frac{2}{x}}}$$

$$c^3 x = 2$$

$$c^3 = \frac{2}{x}$$

$$y = \sqrt[3]{\frac{2x^3}{x}} + \frac{1}{\sqrt[3]{\frac{2}{x}}}$$

$$y = \sqrt[3]{2x^2} + \frac{\sqrt[3]{x}}{\sqrt[3]{2}}$$

$$y = \underline{\hspace{2cm}}$$

$$1) \quad y - x \frac{dy}{dx} = \left(\frac{dy}{dx} \right)^2$$

$$y - xp = p^2$$

$$\frac{dy}{dx} = 2p \frac{dp}{dx} + 1 + 2 \frac{dp}{dx} x$$

$$= \frac{dp}{dx} (2p + 2x)$$

$$p = c$$

$$y = c^2 + 2xc$$

$$2) \quad y + \left(\frac{dy}{dx}\right)^2 = x \frac{dy}{dx}$$

$$y + p^2 = xp$$

$$\frac{dy}{dx} = 1 + 2x \frac{dp}{dx} - 2p \frac{dp}{dx}$$

$$0 = \frac{dp}{dx} (x - 2p)$$

$$p = c$$

$$y = xc - c^2$$

$$3) \quad y = x \frac{dy}{dx} + \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\frac{dy}{dx} = 1 + 2x \frac{dp}{dx} + (\sqrt{1 + p^2})^{-1/2} \cdot 2p \cdot \frac{dp}{dx}$$

$$0 = \frac{dp}{dx} \left(x + (\sqrt{1 + p^2})^{-1/2} 2p \right)$$

$$p = c$$

$$y = xc + \sqrt{1 + c^2}$$

$$4) \quad y = x \frac{dy}{dx} + \frac{dy}{dx}$$

$$1 = 1 + 2x \frac{dp}{dx} + \frac{dp}{dx}$$

$$0 = \frac{dp}{dx} (2x + 1)$$

$$0 = \frac{dy}{dx} (x + y)$$

$$p = C$$

$$y = xc + C$$

$$5) y = x \frac{dy}{dx} + \sin\left(\frac{dy}{dx}\right)$$

$$\frac{dy}{dx} = \cancel{p} + \frac{dp}{dx} x + \cos p \frac{dp}{dx}$$

$$0 = \frac{dp}{dx} (x + \cos p)$$

$$p = C$$

$$y = xc + \sin C$$